Michael Xie

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Education

University of Waterloo *Waterloo, ON*
Bachelor of Computer Science(Co-op) **Expected Graduation:** June 2027

Work Experience

Software Developer Intern May 2026 – August 2026
PlayStation Waterloo, ON
- Incoming Intern within the Infrastructure Team; set to contribute to scalable backend systems and internal tooling.

Software Engineering Intern January 2026 – April 2026
BitGo*- Onboarding Team* Toronto, ON
- Architected automated workflows using **Temporal, Express.js, and PostgreSQL** to optimize the onboarding upgrade flow, resulting in a **23%** increase in operational efficiency.
- Optimized CI/CD pipelines and testing suites using **Vitest, ArgoCD, and Kubernetes**; leveraged **Grafana** for monitoring to achieve a **72%** reduction in test execution time.

Software Developer Intern September 2025 – December 2025
PlayStation*- Payment Parity Team* Waterloo, ON
- Implemented PCI-compliant data masking across **three Spring Boot** orchestration services, reducing sensitive-field exposure and aligning cross-team services under a unified security standard.
- Designed and delivered a consolidated **REST API** endpoint supporting both payment and tax workflows, cutting redundant service calls and simplifying integration for partner teams.

Software Developer Intern January 2025 – April 2025
Royal Bank of Canada*- Digital Development Security Team* Toronto, ON
- Developed an internal AI Assistant using **Langchain**, **RAG**, **PgVector**, and semantic chunking techniques to streamline documentation access, improving response relevance and reducing latency by **30%**.
- Built a **Spring Boot** sidecar service containerized with Docker and deployed on **OpenShift 4** to isolate fraud detection services, reducing duplicate code and simplifying streamlining integration for developers.

Software Developer Intern May 2024 – August 2024
Royal Bank of Canada*- Digital Development Security Team* Toronto, ON
- Built a web tool for decoding JWTs and internal RBC Security Tokens using **Angular** and **Spring Boot**, adopted by **1,000+** employees to speed up token analysis and reduce support requests.
- Reduced onboarding time by **20%** by creating and deploying a token onboarding portal across multiple development environments using **Redis**, **Docker**, and **Openshift 4**.

Technical Systems Analyst Intern May 2023 – August 2023
Royal Bank of Canada*- Law Group Technology Team* Toronto, ON
- Increased productivity by **25%** through automation of Power of Attorney document creation using **ASP.NET**, **Bootstrap** and **jQuery**.

Projects

NES Emulator | *Rust* GitHub
- Designed and implemented a Nintendo Entertainment System(NES) emulator from scratch using **Rust**, achieving accurate hardware emulation and game compatibility.
- Developed a precise emulation of the MOS 6502 microprocessor, replicating original timing and behavior, with built-in tracing to verify instruction accuracy and support debugging.

Home Server Infrastructure | *Proxmox, Linux, Docker, Bash*
- Architected and managed a home lab server using **Proxmox** to run multiple **LXC** containers and VMs for service isolation, microservice testing, and OS experimentation.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, Go, Rust, C/C++, HTML/CSS
Frameworks/Libraries: Spring Boot, Angular, React, OpenCV, Express, Keras, Spring Boot, Node.js, Tailwind CSS
Developer Tools: Git, GitHub Actions, Docker, Grafana, Temporal, ArgoCD, Bash, Nginx, Kubernetes, AWS